

Image Caption Generator

ITAI 1378
GROUP 22600

An AI system for Automatic Image description
Tier 2

TEAM MEMBERS:

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THE PROBLEM

- **WHAT PROBLEM ARE YOU SOLVING?**

- Images do not always have descriptions
- It can be hard to understand image content quickly

- **WHO CARES ABOUT THIS PROBLEM?**

- Social media users
- Visually impaired users
- Content creators

- **WHY IS IT IMPORTANT?**

- Improves accessibility
- Saves time
- Helps organize content



THE SOLUTION

- **What will your system do and how will it solve the problem?**

- An AI system that generates captions for images

- It analyzes the image and describes what is happening

- Automatically outputs a short sentence

- **Benefits:**

- Quick image understanding

- Helps accessibility

- Easy to use

TECHNICAL APPROACH

TECHNIQUE:

- Image captioning

MODEL:

- Pre-trained vision + language model
(like CNN + RNN or BLIP)

FRAMEWORK:

- Python
- PyTorch / Transformers

We chose this approach because it allows the model to understand what is happening in an image and turn it into a description. It combines image recognition and language which makes it useful for generating captions automatically. This helps users quickly understand images without needing manual descriptions.

DATA PLAN

SOURCE:

Kaggle: COCO Image Captioning Dataset
captions_train2017 and captions_val2017

LABELS:

One or more written sentences describing the image.
e.g: "A child in a red jacket"...

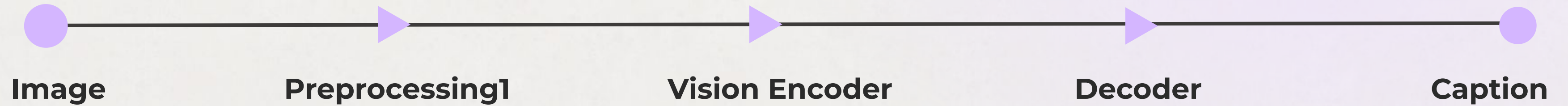
SIZE

600~700 images combining both training and evaluation

PREPARATION:

Cleaning, validating, formatting, preprocessing and splitting the data to prepare for training and evaluation...

SYSTEM DIAGRAM



PRIMARY METRIC

CIDEr is designated as our primary metric because it is purpose-built for image captioning, uses TF-IDF weighting to reward image-specific and discriminative language over generic phrasing, and correlates more strongly with human judgment than BLEU or ROUGE across multiple studies. It is the de facto standard for model comparison on MS-COCO, meaning our results will be directly interpretable alongside published baselines. In practice, CIDEr will be used to select the best model checkpoint during training, determine optimal beam search width, and decide between architectural variants in our ablation studies.

SECONDARY METRICS

BLEU-4 — Included for comparability with published baselines; measures n-gram precision but is limited by its insensitivity to recall and lexical variation.

METEOR — Extends BLEU with recall and synonym matching; a gap between METEOR and BLEU-4 signals healthy lexical generalization.

ROUGE-L — Measures longest common subsequence overlap; flags captions with correct words but poor word order or syntactic structure.

SPICE — Scores object, attribute, and relation accuracy via scene graph parsing; our strongest secondary signal for semantic correctness.

Human Ratings — Serve as qualitative ground truth; any disagreement with CIDEr will be treated as a finding rather than dismissed.

METRICS HIERACHY

Tier	Metric	Role
Primary	CIDEr	Model selection, checkpoint saving, ablation decisions
Secondary (Semantic)	SPICE	Semantic correctness, object/relation accuracy
Secondary (Lexical)	METEOR	Recall and synonym-robust precision
Secondary (Structural)	ROUGE-L	Word order and fluency proxy
Secondary (Legacy)	BLEU-4	Comparability with published baselines
Ground Truth	Human ratings	Final qualitative validation

WEEK-BY-WEEK PLAN

<i>WEEK</i>	<i>TASK</i>	<i>MILESTONE</i>
APRIL 10TH	Set up Github repo, Get dataset and set up environment	Dataset and model ready
APRIL 15TH	Train the model on the images, test captions, fix low accuracy	Model working
April 22nd	Build a simple demo where you upload an image and get a caption, record video	Demo ready
April 28th	Test with different images, write up the documentation	READY
May 6th	Finals	Presentation

CHALLENGES & BACKUP PLANS

One major challenge in an image caption generator is inaccurate or biased captions, especially when the model misinterprets complex or unclear images; a backup plan is to include human review or confidence thresholds before displaying results. Another risk is poor performance on unseen or diverse data, which can be addressed by retraining the model with more diverse datasets or switching to a more advanced pre-trained model.

Latency or slow processing can also be an issue, so a Plan B would be to use a lighter model or process images in batches. Additionally, privacy concerns may arise when handling user-uploaded images, so implementing data encryption and limiting storage is important. Overall, having fallback models, human oversight, and strong data handling policies ensures the system remains reliable and secure.

RESOURCES NEEDED

COMPUTE:

We will use Google Colab and Kaggle for all model training and testing. Both platforms provide free GPU access which is enough to run and fine-tune BLIP-2 without needing a personal machine.

FRAMEWORKS:

PyTorch is the main deep learning framework we will use to build and run our model. Hugging Face Transformers works on top of PyTorch and gives us easy access to pretrained models like BLIP-2, allowing us to load and fine-tune the model in just a few lines of code.

ESTIMATED COST:

This entire project will be built at no cost. All tools, datasets, and compute platforms used are completely free

BLESSING AJUGHU'S CONTRUBTION

- Set up the PowerPoint slides theme and overall design
- Built out the Resources Needed section covering compute, frameworks, APIs, cost, and storage
- Put together the Week-by-Week Plan slide outlining all five milestones from April 10 through May 6
- Reviewed the overall project to make sure everything was accurate and complete

WAYNE LU'S CONTRUBTION

- Researched Primary & secondary metrics
- Built out the primary and secondary metrics sections
- Put together the metric hierarchy for the project

JACQUELINE LOMAS VAZQUEZ'S CONTRIBUTION

- Created Problem, Solution, and Technical Approach sections
- Helped organize project content and structure
- Contributed to presentation slides (Slides 2-4)
- Reviewed and edited project for clarity

ANASHRAH ADIL'S CONTRUBTION

- Contributed created
- Github repository
- Creating the Data plan
- Creating System Design

EMAAN GOHAR'S CONTRUBTION

- Created Challenges and Bakup Plans section
- Helped organize project content and structure
- Reviewed project to ensure everything was accurate and complete